PRESENTATION NOTES

# OC Venue Activation Knowledge Base

What changed across Phases 1, 2, and 3 — and what's still left to do. A knowledge base that joins Fullerton's permit rules, fee schedules, and market-rate data into one cited answer, instead of three separate lookups.

Build progress summary · August 2026

## The One-Line Pitch

Someone asking "can I host this activation, and will it pencil out" is really asking three questions at once — is it legal, is it profitable, is it affordable — and today those three answers live in three different places: a city's code, a permit fee PDF, and a venue marketplace. This project puts all three in one place, with every claim traceable back to its source, and an explicit refusal instead of a guess whenever the data isn't there.

> **SCOPE, RIGHT NOW**
>
> Fullerton only. Wedding venues and gyms hosting one-day "activations." Three phases in, still building toward the full nine-city Central Orange County plan.

## Phase 1 — Prove the Pipeline

> *Goal: one source type, ingested end to end, with search that returns real cited results and something actually deployed.*

| 4 | 35 | 1 |
|---|---|---|
| DOCUMENTS | CHUNKS | INGESTION METHOD |

Fullerton's municipal code chapters on special events and Conditional Use Permits, plus the actual fee schedule, went into Supabase with a vector database (pgvector) behind it. Every chunk carries its source section and a citation — that discipline started here and never got relaxed.

### The one technical idea worth explaining out loud

**Hybrid search**: two search methods running together. Keyword search catches exact things — "$100," "Chapter 15.58" — that a meaning-based search tends to blur. Meaning-based (semantic) search catches paraphrases a keyword search would miss entirely — "temporary event" matching "special event" even with no shared words. Running both and combining the results covers what either one misses alone.

## Phase 2 — Widen the Intake

> *Goal: all three required ingestion methods working — an API, a live website, a document upload — plus safe re-ingestion and a way to see what's actually in the system.*

| 50 | 89 | 3 |
|---|---|---|
| DOCUMENTS | CHUNKS | INGESTION METHODS |

### What actually got ingested

- **API:** 43 real Fullerton venues and gyms, pulled from Google Places.
- **Live website:** OC Fire Authority's special-event permit page — directly answers the "does this trigger a fire-marshal review" question.

- **Document upload:** California's statewide alcohol-permit reference forms (two different permit paths, with real dollar figures verified against the primary government PDFs, not just a search summary).

> **THE HONEST PIVOT**
>
> The original plan for the API and one live-website source (Yelp, Eventbrite, and a venue marketplace called Peerspace) fell apart on contact with reality — paywalled, retired, or blocking automated access. That's not a failure to hide; it's exactly the "sources are messy" lesson this kind of project is supposed to teach. All three got swapped for sources that actually work, and the swap is documented, not papered over.

Also caught and fixed a real bug in our own pipeline: re-running ingestion had silently created a duplicate document. Fixed with a proper database constraint plus an "update, don't duplicate" pattern — now ingestion can be re-run safely any time a source changes.

Deployed live on Railway at this point, with a page for browsing every ingested source directly.

## Phase 3 — Sharpen It (complete)

> *Goal: move from "here are some matching passages" to a real chat interface that gives a grounded, cited answer — and knows when to say "I don't know."*

| 57 | 96 | 13 |
|---|---|---|
| DOCUMENTS | CHUNKS | TEST QUESTIONS GRADED |

### Step one: measure honestly before changing anything

All 13 of the original test questions were run against the system exactly as Phase 2 left it — no generation layer yet, just raw search results. Graded openly: **3 good, 5 partial, 5 outright failing.** That score is the baseline everything since gets measured against.

> **THE MOST INTERESTING FINDING**
>
> Two questions should have been refused outright — one asking for investment advice, one asking for a revenue forecast. They failed for two *different* reasons. One scored close to this system's "nothing relevant" floor, so a simple confidence cutoff could have caught it. The other scored well because it happened to mention real matching content — the *content* was relevant even though the *question type* wasn't. No confidence cutoff can ever catch that second case; it needs the system to actually understand what kind of question is being asked.

### What got built in response

- A real chat layer (Google Gemini) that reads the retrieved passages and either answers with citations, or says plainly "this isn't in the knowledge base" — instead of quietly presenting a loosely-related passage as if it were the answer.
- An explicit check for the two out-of-scope categories, applied before search even runs.
- Real market-rate data: seven actual Fullerton venue prices, individually verified to make sure venues from neighboring cities didn't sneak in through a marketplace site's search radius. This closed a specific, previously-flagged gap (no real pricing existed before this).

Re-tested against the original failing questions: the two "should have refused" questions now correctly decline. The two "confidently wrong" questions — where the system had been presenting the right-looking-but-wrong-source data as if it answered the question — now correctly say the information isn't available instead.

### Closing the loop: reranking and an agentic-retrieval experiment

Two techniques were added, targeting two genuinely different failure modes rather than one fix wearing two names. **Reranking** widens the search to 15 candidates and has the model score each one for real relevance to the actual question, not just embedding similarity — this is what catches a passage from the wrong city, permit type, or jurisdiction even when it scores deceptively high on similarity alone. **Agentic retrieval** — the brief's required Phase 3 experiment — kicks in only when reranking finds nothing genuinely relevant: the model rewrites the search query itself and searches again, capped at one extra hop. This is what actually fixed the gym-permit case flagged above: the right permit chapter existed in the corpus the whole time, it just never matched that exact phrasing until the system rewrote the query itself.

| 0 | 2 | 9 |
|---|---|---|
| FAIL (WAS 5) | ANSWERED W/ REAL DATA | HONEST REFUSALS |

Re-running the full 13-question set through the finished chat layer — the real before/after number the baseline was always waiting on — the most dangerous baseline failure is gone outright: a different jurisdiction's permit fee (from a state alcohol form) is no longer presented as if it were Fullerton's own threshold, at 0.876 similarity, "dangerously close to plausible." It's now a clean, honest refusal instead. Full grading in the evaluation writeup.

## What's Left

### Known, honestly-flagged gaps

- Fullerton's insurance/certificate-of-insurance requirements — never found published anywhere by the city itself. Left out on purpose rather than guessing a generic California number.

### Documentation still to assemble

- The formal retrospective.
- A demo script / screenshot set for handover.

### Deferred, not forgotten

Phase 4 — the full nine-city expansion — is intentionally not started. One design decision from this phase is already banked for when it does start: city disambiguation has to be a metadata filter applied before search ranking, not something left to the search itself to figure out, because near-identical fee language across nine cities makes that kind of mix-up an easy, real risk.

*"Build something that finds the right answer, not just an answer." Three phases in, the system is getting measurably better at knowing the difference between the two.*